

Abstract Missed fractures remain one of the most frequent sources of diagnostic errors in emergency departments, often leading to delayed treatment, morbidity, and increased healthcare costs. Artificial intelligence (AI), particularly deep learning systems, has been increasingly investigated as an adjunct for musculoskeletal imaging. Over the past five years, multiple studies have evaluated the diagnostic performance, clinical utility, and limitations of AI in fracture detection. This article is a narrative synthesis of literature published from 2020 to 2025, focusing on systematic reviews, meta-analyses, and high-quality prospective studies addressing AI-assisted fracture diagnosis on radiographs and other imaging modalities. Key themes examined include diagnostic accuracy, anatomical and modality-specific performance, real-world deployment, regulatory approvals, and remaining challenges to integration. The results of the current review showed that decent meta-analyses demonstrated pooled sensitivity and specificity above 90%, showing AI performance comparable to that of radiologists. The strongest results were observed in extremity fractures (wrist, ankle, shoulder), while performance was more variable in ribs and spine. Reader studies confirmed that AI assistance improved radiologists' sensitivity by 6-8% without loss of specificity. Real-world deployments in emergency departments showed modest reductions in reporting discrepancies and patient length of stay. Commercial platforms, such as OsteoDetect and BoneView, received U.S. Food and Drug Administration (FDA) and Conformité Européenne (CE) approvals, and draft guidance from the National Institute for Health and Care Excellence (NICE) in 2024 recommended AI fracture detection in urgent care. However, challenges persist, including dataset bias, limited generalizability, interpretability, and uncertain patient-centered outcomes. Several studies have reported that AI may serve as a reliable adjunct in fracture diagnosis, with diagnostic accuracy approaching that of radiologists and potential improvements in clinical workflows when used as an assistive tool. Nevertheless, safe integration is best supported by regulatory approval (FDA, CE, NICE) and, where appropriate, supplemented by external validation, robust datasets, and transparent reporting to ensure performance across different clinical environments. Future directions include multi-modal AI, adaptive learning, pediatric-focused applications, and strengthened real-world

evidence from retrospective outcome evaluations and post-market surveillance.